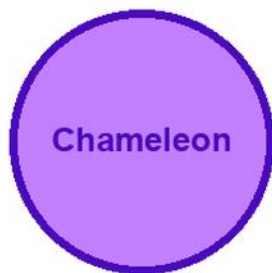


KITABU QUEST

S4

CHEMISTRY



Cover scene

students doing safe chemistry work with a chameleon mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

4

Title Page

KITABU QUEST S4 Chemistry

Rwanda REB-CBC learner book

Mascot: chameleon

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S4 learners study Chemistry one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use safe observations, simple models, data tables, and evidence-based explanations.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Structure of an atom and mass spectrum
2. Electron configurations of atoms and ions
3. Formation of ionic and metallic bonds
4. Covalent bond and molecular structures
5. Variation in trends of the physical properties
6. Trends in chemical properties of Group 1 elements and their compounds
7. Trends in chemical properties of Group 2 elements and their compounds
8. Trends of chemical properties of Group 13 elements and their compounds
9. Trends in chemical properties of Group 14 elements and their compounds
10. Trends in chemical properties of Group 15 elements and their compounds
11. Trends of chemical properties of Group 16 elements and their compounds
12. Trends of chemical properties of Group 17 elements and their compounds
13. Properties and uses of Group 18 elements
14. Trends in chemical properties of
15. Factors that affect chemical equilibrium
16. Acids and bases
17. Reduction and oxidation reactions
18. Energy changes and energy profile diagrams for chemical reactions

As you finish each topic, return to the success criteria and correct one answer before moving on.

Structure of an atom and mass spectrum: Lesson Opener

Learning focus: Structure of an atom and mass spectrum

Country link: a safe Rwandan observation, model, data table, or investigation for Structure of an atom and mass spectrum.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the interpret simple mass spectra and use them to calculate R.A.M. of different elements
- outline the discovery of the sub-atomic particles
- compare the properties of sub- atomic particles
- explain the fundamental processes that occur in the functioning of a mass spectrometer
- state the uses of the mass spectrometer
- interpret different mass spectra

Inquiry questions:

- How can structure of an atom and mass spectrum help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Structure of an atom and mass spectrum: Learn

Structure of an atom and mass spectrum is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Structure of an atom and mass spectrum
- Structure
- atom
- mass
- spectrum
- concept
- observation
- evidence

Structure of an atom and mass spectrum: Worked Example

Worked example for Structure of an atom and mass spectrum: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Structure of an atom and mass spectrum: Vocabulary And Study Skill

Use these words and study moves before you practise Structure of an atom and mass spectrum. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Structure of an atom and mass spectrum
- Structure
- atom
- mass
- spectrum
- concept

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Structure of an atom and mass spectrum without a clear example, method, or evidence.

Structure of an atom and mass spectrum: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Structure of an atom and mass spectrum.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Structure of an atom and mass spectrum: Practice And Correction

Practice for Structure of an atom and mass spectrum:

Main outcome: the interpret simple mass spectra and use them to calculate R.A.M. of different elements.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Structure of an atom and mass spectrum: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Structure of an atom and mass spectrum.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Structure of an atom and mass spectrum: Outcome Check

Outcome check for Structure of an atom and mass spectrum: The interpret simple mass spectra and use them to calculate R.A.M. of different elements.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Structure of an atom and mass spectrum: Outcome Check

Outcome check for Structure of an atom and mass spectrum: Outline the discovery of the sub-atomic particles.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electron configurations of atoms and ions: Lesson Opener

Learning focus: Electron configurations of atoms and ions

Country link: a safe Rwandan observation, model, data table, or investigation for Electron configurations of atoms and ions.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the relate Bohr's model of the atom with hydrogen spectrum and energy levels, practice writing electronic configurations using s, p, d, f orbitals and interpret graphical information in relation to ionization energy of elements
- explain how the data from emission spectra provide evidence for discrete energy levels within the atom
- describe the atomic orbitals
- determine the electronic configurations of the atoms and ions in terms of s, p, d and f orbitals
- derive the electronic configuration of an element from data on successive ionization energies
- simple interpretation of spectral line series of hydrogen atom

Inquiry questions:

- How can electron configurations of atoms and ions help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electron configurations of atoms and ions: Learn

Electron configurations of atoms and ions is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Electron configurations of atoms and ions
- Electron
- configurations
- atoms
- ions
- concept
- observation
- evidence

Electron configurations of atoms and ions: Worked Example

Investigation model for Electron configurations of atoms and ions: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Electron configurations of atoms and ions: Vocabulary And Study Skill

Use these words and study moves before you practise Electron configurations of atoms and ions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Electron configurations of atoms and ions
- Electron
- configurations
- atoms
- ions
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Electron configurations of atoms and ions without a clear example, method, or evidence.

Electron configurations of atoms and ions: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Electron configurations of atoms and ions.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Electron configurations of atoms and ions: Practice And Correction

Practice for Electron configurations of atoms and ions:

Main outcome: the relate Bohr's model of the atom with hydrogen spectrum and energy levels, practice writing electronic configurations using s, p, d, f orbitals and interpret graphical information in relation to ionization energy of elements.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Electron configurations of atoms and ions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Electron configurations of atoms and ions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electron configurations of atoms and ions: Outcome Check

Outcome check for Electron configurations of atoms and ions: The relate Bohr's model of the atom with hydrogen spectrum and energy levels, practice writing electronic configurations using s, p, d, f orbitals and interpret graphical information in relation to ionization energy of elements.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electron configurations of atoms and ions: Outcome Check

Outcome check for Electron configurations of atoms and ions: Explain how the data from emission spectra provide evidence for discrete energy levels within the atom.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Formation of ionic and metallic bonds: Lesson Opener

Learning focus: Formation of ionic and metallic bonds

Country link: a safe Rwandan observation, model, data table, or investigation for Formation of ionic and metallic bonds.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the describe how properties of ionic compounds and metals are related to the nature of their bonding
- explain the mechanisms by which atoms of different elements attain stability
- explain the formation of ionic bonds
- describe the properties of ionic compounds
- state the factors that influence the magnitude of lattice energy
- describe the formation of metallic bonds

Inquiry questions:

- How can formation of ionic and metallic bonds help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Formation of ionic and metallic bonds: Learn

Formation of ionic and metallic bonds explains why atoms join and how bonds affect properties.

Core idea:

- Ionic bonding happens when electrons transfer, often between metals and non-metals.
- Covalent bonding happens when atoms share electrons, usually between non-metals.
- Metallic bonding has metal ions in a sea of delocalized electrons.
- Bonding helps explain melting point, conductivity, solubility, and hardness.
- Dot-and-cross or particle diagrams make bonding easier to compare.

Key words:

- Formation of ionic and metallic bonds
- Formation
- ionic
- metallic
- bonds
- concept
- observation
- evidence

Formation of ionic and metallic bonds: Worked Example

Model for Formation of ionic and metallic bonds: Compare sodium chloride and a simple covalent molecule.

1. Sodium transfers one electron to chlorine, forming ions.
2. Opposite charges attract, so an ionic bond forms.
3. In a covalent molecule, atoms share electrons instead of transferring them.
4. Ionic compounds often conduct electricity when molten or dissolved; simple covalent substances usually do not.

Conclusion: the type of bond helps explain properties.

Formation of ionic and metallic bonds: Vocabulary And Study Skill

Use these words and study moves before you practise Formation of ionic and metallic bonds. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Formation of ionic and metallic bonds
- Formation
- ionic
- metallic
- bonds
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Formation of ionic and metallic bonds without a clear example, method, or evidence.

Formation of ionic and metallic bonds: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Formation of ionic and metallic bonds.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Formation of ionic and metallic bonds: Practice And Correction

Practice for Formation of ionic and metallic bonds:

Main outcome: the describe how properties of ionic compounds and metals are related to the nature of their bonding.

1. Complete a table comparing ionic, covalent, and metallic bonding.
2. Draw a simple electron-transfer or sharing model.
3. Match each bond type with one property such as conductivity, melting point, or hardness.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Formation of ionic and metallic bonds: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Formation of ionic and metallic bonds.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Formation of ionic and metallic bonds: Outcome Check

Outcome check for Formation of ionic and metallic bonds: The describe how properties of ionic compounds and metals are related to the nature of their bonding.

1. Compare ionic and covalent bonding in one table.
2. State whether electrons are transferred or shared.
3. Link one bond type to one property.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Formation of ionic and metallic bonds: Outcome Check

Outcome check for Formation of ionic and metallic bonds: Explain the mechanisms by which atoms of different elements attain stability.

1. Compare ionic and covalent bonding in one table.
2. State whether electrons are transferred or shared.
3. Link one bond type to one property.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Covalent bond and molecular structures: Lesson Opener

Learning focus: Covalent bond and molecular structures

Country link: a safe Rwandan observation, model, data table, or investigation for Covalent bond and molecular structures.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the demonstrate how the nature of the bonding is related to the properties of covalent compounds and molecular structures
- define octet rule as applied to covalent compounds
- explain the formation of covalent bonds and describe the properties of covalent compounds
- describe how the properties of covalent compounds depend on their bonding
- explain the VSEPR theory
- describe simple and giant covalent molecular structures

Inquiry questions:

- How can covalent bond and molecular structures help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Covalent bond and molecular structures: Learn

Covalent bond and molecular structures explains why atoms join and how bonds affect properties.

Core idea:

- Ionic bonding happens when electrons transfer, often between metals and non-metals.
- Covalent bonding happens when atoms share electrons, usually between non-metals.
- Metallic bonding has metal ions in a sea of delocalized electrons.
- Bonding helps explain melting point, conductivity, solubility, and hardness.
- Dot-and-cross or particle diagrams make bonding easier to compare.

Key words:

- Covalent bond and molecular structures
- Covalent
- bond
- molecular
- structures
- concept
- observation
- evidence

Covalent bond and molecular structures: Worked Example

Model for Covalent bond and molecular structures: Compare sodium chloride and a simple covalent molecule.

1. Sodium transfers one electron to chlorine, forming ions.
2. Opposite charges attract, so an ionic bond forms.
3. In a covalent molecule, atoms share electrons instead of transferring them.
4. Ionic compounds often conduct electricity when molten or dissolved; simple covalent substances usually do not.

Conclusion: the type of bond helps explain properties.

Covalent bond and molecular structures: Vocabulary And Study Skill

Use these words and study moves before you practise Covalent bond and molecular structures. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Covalent bond and molecular structures
- Covalent
- bond
- molecular
- structures
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Covalent bond and molecular structures without a clear example, method, or evidence.

Covalent bond and molecular structures: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Covalent bond and molecular structures.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Covalent bond and molecular structures: Practice And Correction

Practice for Covalent bond and molecular structures:

Main outcome: the demonstrate how the nature of the bonding is related to the properties of covalent compounds and molecular structures.

1. Complete a table comparing ionic, covalent, and metallic bonding.
2. Draw a simple electron-transfer or sharing model.
3. Match each bond type with one property such as conductivity, melting point, or hardness.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Covalent bond and molecular structures: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Covalent bond and molecular structures.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Covalent bond and molecular structures: Outcome Check

Outcome check for Covalent bond and molecular structures: The demonstrate how the nature of the bonding is related to the properties of covalent compounds and molecular structures.

1. Compare ionic and covalent bonding in one table.
2. State whether electrons are transferred or shared.
3. Link one bond type to one property.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Covalent bond and molecular structures: Outcome Check

Outcome check for Covalent bond and molecular structures: Define octet rule as applied to covalent compounds.

1. Compare ionic and covalent bonding in one table.
2. State whether electrons are transferred or shared.
3. Link one bond type to one property.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Variation in trends of the physical properties: Lesson Opener

Learning focus: Variation in trends of the physical properties

Country link: a safe Rwandan observation, model, data table, or investigation for Variation in trends of the physical properties.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the use atomic structure and electronic configuration to explain the trends in the physical properties of the elements
- outline the historical back ground of the Periodic Table
- explain the trends in the physical properties of the elements across a period and down a group
- classify the elements into respective groups and periods using electronic configuration
- relate trends in physical properties of the elements to their electronic configuration
- classify the elements into blocks (s, p, d, f- block)

Inquiry questions:

- How can variation in trends of the physical properties help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Variation in trends of the physical properties: Learn

Variation in trends of the physical properties is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Variation in trends of the physical properties
- Variation
- trends
- physical
- properties
- concept
- observation
- evidence

Variation in trends of the physical properties: Worked Example

Investigation model for Variation in trends of the physical properties: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Variation in trends of the physical properties: Vocabulary And Study Skill

Use these words and study moves before you practise Variation in trends of the physical properties. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Variation in trends of the physical properties
- Variation
- trends
- physical
- properties
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Variation in trends of the physical properties without a clear example, method, or evidence.

Variation in trends of the physical properties: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Variation in trends of the physical properties.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Variation in trends of the physical properties: Practice And Correction

Practice for Variation in trends of the physical properties:

Main outcome: the use atomic structure and electronic configuration to explain the trends in the physical properties of the elements.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Variation in trends of the physical properties: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Variation in trends of the physical properties.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Variation in trends of the physical properties: Outcome Check

Outcome check for Variation in trends of the physical properties: The use atomic structure and electronic configuration to explain the trends in the physical properties of the elements.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Variation in trends of the physical properties: Outcome Check

Outcome check for Variation in trends of the physical properties: Outline the historical back ground of the Periodic Table.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 1 elements and their compounds: Lesson Opener

Learning focus: Trends in chemical properties of Group 1 elements and their compounds

Country link: a safe Rwandan observation, model, data table, or investigation for Trends in chemical properties of Group 1 elements and their compounds.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the learner should be able to compare and contrast the chemical properties of the Group 1 elements and their compounds in relation to their position in the Periodic Table
- describe and explain the physical properties of Group 1 elements in terms of metallic character and strength of metallic bond
- describe and explain the reactivity of the Group 1 elements with oxygen, water and the halogens
- state and explain the properties of Group 1 oxides and hydroxides
- explain the trends in the solubility of Group 1 compounds

Inquiry questions:

- How can trends in chemical properties of group 1 elements and their compounds help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 1 elements and their compounds: Learn

Trends in chemical properties of Group 1 elements and their compounds explains why atoms join and how bonds affect properties.

Core idea:

- Ionic bonding happens when electrons transfer, often between metals and non-metals.
- Covalent bonding happens when atoms share electrons, usually between non-metals.
- Metallic bonding has metal ions in a sea of delocalized electrons.
- Bonding helps explain melting point, conductivity, solubility, and hardness.
- Dot-and-cross or particle diagrams make bonding easier to compare.

Key words:

- Trends in chemical properties of Group 1 elements and their compounds
- Trends
- chemical
- properties
- Group
- elements
- their
- compounds

Trends in chemical properties of Group 1 elements and their compounds: Worked Example

Model for Trends in chemical properties of Group 1 elements and their compounds: Compare sodium chloride and a simple covalent molecule.

1. Sodium transfers one electron to chlorine, forming ions.
2. Opposite charges attract, so an ionic bond forms.
3. In a covalent molecule, atoms share electrons instead of transferring them.
4. Ionic compounds often conduct electricity when molten or dissolved; simple covalent substances usually do not.

Conclusion: the type of bond helps explain properties.

Trends in chemical properties of Group 1 elements and their compounds: Vocabulary And Study Skill

Use these words and study moves before you practise Trends in chemical properties of Group 1 elements and their compounds. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Trends in chemical properties of Group 1 elements and their compounds
- Trends
- chemical
- properties
- Group
- elements
- their
- compounds

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Trends in chemical properties of Group 1 elements and their compounds without a clear example, method, or evidence.

Trends in chemical properties of Group 1 elements and their compounds: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Trends in chemical properties of Group 1 elements and their compounds.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Trends in chemical properties of Group 1 elements and their compounds: Practice And Correction

Practice for Trends in chemical properties of Group 1 elements and their compounds:

Main outcome: the learner should be able to compare and contrast the chemical properties of the Group 1 elements and their compounds in relation to their position in the Periodic Table.

1. Complete a table comparing ionic, covalent, and metallic bonding.
2. Draw a simple electron-transfer or sharing model.
3. Match each bond type with one property such as conductivity, melting point, or hardness.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Trends in chemical properties of Group 1 elements and their compounds: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Trends in chemical properties of Group 1 elements and their compounds.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 1 elements and their compounds: Outcome Check

Outcome check for Trends in chemical properties of Group 1 elements and their compounds: The learner should be able to compare and contrast the chemical properties of the Group 1 elements and their compounds in relation to their position in the Periodic Table.

1. Compare ionic and covalent bonding in one table.
2. State whether electrons are transferred or shared.
3. Link one bond type to one property.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 1 elements and their compounds: Outcome Check

Outcome check for Trends in chemical properties of Group 1 elements and their compounds: Describe and explain the physical properties of Group 1 elements in terms of metallic character and strength of metallic bond.

1. Compare ionic and covalent bonding in one table.
2. State whether electrons are transferred or shared.
3. Link one bond type to one property.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 2 elements and their compounds: Lesson Opener

Learning focus: Trends in chemical properties of Group 2 elements and their compounds

Country link: a safe Rwandan observation, model, data table, or investigation for Trends in chemical properties of Group 2 elements and their compounds.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the compare and contrast the properties of the Group 2 elements and their compounds in relation to their position in the Periodic Table
- describe the physical properties of Group 2 elements
- describe the properties of Group 2 oxides and hydroxides
- explain the trends in the thermal decomposition of Group 2 carbonates, and nitrates
- explain the trends in the solubility of Group 2 compounds
- state the uses of Group 2 elements and their compounds

Inquiry questions:

- How can trends in chemical properties of group 2 elements and their compounds help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 2 elements and their compounds: Learn

Trends in chemical properties of Group 2 elements and their compounds is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Trends in chemical properties of Group 2 elements and their compounds
- Trends
- chemical
- properties
- Group
- elements
- their
- compounds

Trends in chemical properties of Group 2 elements and their compounds: Worked Example

Investigation model for Trends in chemical properties of Group 2 elements and their compounds:
Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Trends in chemical properties of Group 2 elements and their compounds: Vocabulary And Study Skill

Use these words and study moves before you practise Trends in chemical properties of Group 2 elements and their compounds. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Trends in chemical properties of Group 2 elements and their compounds
- Trends
- chemical
- properties
- Group
- elements
- their
- compounds

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Trends in chemical properties of Group 2 elements and their compounds without a clear example, method, or evidence.

Trends in chemical properties of Group 2 elements and their compounds: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Trends in chemical properties of Group 2 elements and their compounds.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Trends in chemical properties of Group 2 elements and their compounds: Practice And Correction

Practice for Trends in chemical properties of Group 2 elements and their compounds:

Main outcome: the compare and contrast the properties of the Group 2 elements and their compounds in relation to their position in the Periodic Table.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Trends in chemical properties of Group 2 elements and their compounds: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Trends in chemical properties of Group 2 elements and their compounds.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 2 elements and their compounds: Outcome Check

Outcome check for Trends in chemical properties of Group 2 elements and their compounds: The compare and contrast the properties of the Group 2 elements and their compounds in relation to their position in the Periodic Table.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 2 elements and their compounds: Outcome Check

Outcome check for Trends in chemical properties of Group 2 elements and their compounds: Describe the physical properties of Group 2 elements.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends of chemical properties of Group 13 elements and their compounds: Lesson Opener

Learning focus: Trends of chemical properties of Group 13 elements and their compounds

Country link: a safe Rwandan observation, model, data table, or investigation for Trends of chemical properties of Group 13 elements and their compounds.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the compare and contrast the chemical properties of the Group 13 elements and their compounds in relation to their position in the Periodic Table
- state the physical properties of Group 13 elements
- explain the reactivity of Group 13 elements with oxygen, water, halogens, dilute acids and sodium hydroxide
- describe the properties of oxides, hydroxides and chlorides of Group 13 elements
- state the uses of Group 13 elements and their compounds
- compare and contrast the reactivity of Group 13 elements with oxygen, water, halogens, dilute acids and sodium hydroxide

Inquiry questions:

- How can trends of chemical properties of group 13 elements and their compounds help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends of chemical properties of Group 13 elements and their compounds: Learn

Trends of chemical properties of Group 13 elements and their compounds is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Trends of chemical properties of Group 13 elements and their compounds
- Trends
- chemical
- properties
- Group
- elements
- their
- compounds

Trends of chemical properties of Group 13 elements and their compounds: Worked Example

Investigation model for Trends of chemical properties of Group 13 elements and their compounds:
Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Trends of chemical properties of Group 13 elements and their compounds: Vocabulary And Study Skill

Use these words and study moves before you practise Trends of chemical properties of Group 13 elements and their compounds. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Trends of chemical properties of Group 13 elements and their compounds
- Trends
- chemical
- properties
- Group
- elements
- their
- compounds

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Trends of chemical properties of Group 13 elements and their compounds without a clear example, method, or evidence.

Trends of chemical properties of Group 13 elements and their compounds: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Trends of chemical properties of Group 13 elements and their compounds.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Trends of chemical properties of Group 13 elements and their compounds: Practice And Correction

Practice for Trends of chemical properties of Group 13 elements and their compounds:

Main outcome: the compare and contrast the chemical properties of the Group 13 elements and their compounds in relation to their position in the Periodic Table.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Trends of chemical properties of Group 13 elements and their compounds: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Trends of chemical properties of Group 13 elements and their compounds.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends of chemical properties of Group 13 elements and their compounds: Outcome Check

Outcome check for Trends of chemical properties of Group 13 elements and their compounds: The compare and contrast the chemical properties of the Group 13 elements and their compounds in relation to their position in the Periodic Table.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends of chemical properties of Group 13 elements and their compounds: Outcome Check

Outcome check for Trends of chemical properties of Group 13 elements and their compounds: State the physical properties of Group 13 elements.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 14 elements and their compounds: Lesson Opener

Learning focus: Trends in chemical properties of Group 14 elements and their compounds

Country link: a safe Rwandan observation, model, data table, or investigation for Trends in chemical properties of Group 14 elements and their compounds.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the compare and contrast the chemical properties of the Group 14 elements and their compounds in relation to their position in the Periodic Table
- state physical properties of Group 14 elements
- state the chemical properties of Group 14 elements
- distinguish between the chemical reactions of the oxides and chlorides of Group 14 elements
- explain the trends in thermal stability of the oxide, halides and hydrides of Group 14 elements
- explain the variation in stability of oxidation state of +2 and +4 down the

Inquiry questions:

- How can trends in chemical properties of group 14 elements and their compounds help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 14 elements and their compounds: Learn

Trends in chemical properties of Group 14 elements and their compounds is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Trends in chemical properties of Group 14 elements and their compounds
- Trends
- chemical
- properties
- Group
- elements
- their
- compounds

Trends in chemical properties of Group 14 elements and their compounds: Worked Example

Investigation model for Trends in chemical properties of Group 14 elements and their compounds:
Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Trends in chemical properties of Group 14 elements and their compounds: Vocabulary And Study Skill

Use these words and study moves before you practise Trends in chemical properties of Group 14 elements and their compounds. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Trends in chemical properties of Group 14 elements and their compounds
- Trends
- chemical
- properties
- Group
- elements
- their
- compounds

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Trends in chemical properties of Group 14 elements and their compounds without a clear example, method, or evidence.

Trends in chemical properties of Group 14 elements and their compounds: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Trends in chemical properties of Group 14 elements and their compounds.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Trends in chemical properties of Group 14 elements and their compounds: Practice And Correction

Practice for Trends in chemical properties of Group 14 elements and their compounds:

Main outcome: the compare and contrast the chemical properties of the Group 14 elements and their compounds in relation to their position in the Periodic Table.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Trends in chemical properties of Group 14 elements and their compounds: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Trends in chemical properties of Group 14 elements and their compounds.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 14 elements and their compounds: Outcome Check

Outcome check for Trends in chemical properties of Group 14 elements and their compounds: The compare and contrast the chemical properties of the Group 14 elements and their compounds in relation to their position in the Periodic Table.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 14 elements and their compounds: Outcome Check

Outcome check for Trends in chemical properties of Group 14 elements and their compounds: State physical properties of Group 14 elements.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 15 elements and their compounds: Lesson Opener

Learning focus: Trends in chemical properties of Group 15 elements and their compounds

Country link: a safe Rwandan observation, model, data table, or investigation for Trends in chemical properties of Group 15 elements and their compounds.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the learner should be able to compare and contrast the properties of the Group 15 elements and their compounds in relation to their position in the Periodic Table
- describe the physical properties of the Group 15 elements
- describe the variation in the metallic and non-metallic character of the Group 15 elements
- recall the physical properties of the allotropes of phosphorus
- describe the chemical reactions of nitrogen compounds
- be aware of the increasing oxidizing power of nitric acid

Inquiry questions:

- How can trends in chemical properties of group 15 elements and their compounds help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 15 elements and their compounds: Learn

Trends in chemical properties of Group 15 elements and their compounds explains why atoms join and how bonds affect properties.

Core idea:

- Ionic bonding happens when electrons transfer, often between metals and non-metals.
- Covalent bonding happens when atoms share electrons, usually between non-metals.
- Metallic bonding has metal ions in a sea of delocalized electrons.
- Bonding helps explain melting point, conductivity, solubility, and hardness.
- Dot-and-cross or particle diagrams make bonding easier to compare.

Key words:

- Trends in chemical properties of Group 15 elements and their compounds
- Trends
- chemical
- properties
- Group
- elements
- their
- compounds

Trends in chemical properties of Group 15 elements and their compounds: Worked Example

Model for Trends in chemical properties of Group 15 elements and their compounds: Compare sodium chloride and a simple covalent molecule.

1. Sodium transfers one electron to chlorine, forming ions.
2. Opposite charges attract, so an ionic bond forms.
3. In a covalent molecule, atoms share electrons instead of transferring them.
4. Ionic compounds often conduct electricity when molten or dissolved; simple covalent substances usually do not.

Conclusion: the type of bond helps explain properties.

Trends in chemical properties of Group 15 elements and their compounds: Vocabulary And Study Skill

Use these words and study moves before you practise Trends in chemical properties of Group 15 elements and their compounds. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Trends in chemical properties of Group 15 elements and their compounds
- Trends
- chemical
- properties
- Group
- elements
- their
- compounds

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Trends in chemical properties of Group 15 elements and their compounds without a clear example, method, or evidence.

Trends in chemical properties of Group 15 elements and their compounds: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Trends in chemical properties of Group 15 elements and their compounds.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Trends in chemical properties of Group 15 elements and their compounds: Practice And Correction

Practice for Trends in chemical properties of Group 15 elements and their compounds:

Main outcome: the learner should be able to compare and contrast the properties of the Group 15 elements and their compounds in relation to their position in the Periodic Table.

1. Complete a table comparing ionic, covalent, and metallic bonding.
2. Draw a simple electron-transfer or sharing model.
3. Match each bond type with one property such as conductivity, melting point, or hardness.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Trends in chemical properties of Group 15 elements and their compounds: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Trends in chemical properties of Group 15 elements and their compounds.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 15 elements and their compounds: Outcome Check

Outcome check for Trends in chemical properties of Group 15 elements and their compounds: The learner should be able to compare and contrast the properties of the Group 15 elements and their compounds in relation to their position in the Periodic Table.

1. Compare ionic and covalent bonding in one table.
2. State whether electrons are transferred or shared.
3. Link one bond type to one property.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of Group 15 elements and their compounds: Outcome Check

Outcome check for Trends in chemical properties of Group 15 elements and their compounds: Describe the physical properties of the Group 15 elements.

1. Compare ionic and covalent bonding in one table.
2. State whether electrons are transferred or shared.
3. Link one bond type to one property.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends of chemical properties of Group 16 elements and their compounds: Lesson Opener

Learning focus: Trends of chemical properties of Group 16 elements and their compounds

Country link: a safe Rwandan observation, model, data table, or investigation for Trends of chemical properties of Group 16 elements and their compounds.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the learner should be able to compare and contrast the chemical properties of the Group 16 elements and their compounds in relation to their position in the Periodic Table
- describe the physical properties of the Group 16 elements
- describe the reactions between sulphur and oxygen
- describe the steps and conditions applied in the industrial preparations of sulphuric acid
- describe the chemical properties of sulphuric acid
- describe the properties of oxoanions

Inquiry questions:

- How can trends of chemical properties of group 16 elements and their compounds help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends of chemical properties of Group 16 elements and their compounds: Learn

Trends of chemical properties of Group 16 elements and their compounds is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Trends of chemical properties of Group 16 elements and their compounds
- Trends
- chemical
- properties
- Group
- elements
- their
- compounds

Trends of chemical properties of Group 16 elements and their compounds: Worked Example

Investigation model for Trends of chemical properties of Group 16 elements and their compounds:
Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Trends of chemical properties of Group 16 elements and their compounds: Vocabulary And Study Skill

Use these words and study moves before you practise Trends of chemical properties of Group 16 elements and their compounds. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Trends of chemical properties of Group 16 elements and their compounds
- Trends
- chemical
- properties
- Group
- elements
- their
- compounds

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Trends of chemical properties of Group 16 elements and their compounds without a clear example, method, or evidence.

Trends of chemical properties of Group 16 elements and their compounds: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Trends of chemical properties of Group 16 elements and their compounds.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Trends of chemical properties of Group 16 elements and their compounds: Practice And Correction

Practice for Trends of chemical properties of Group 16 elements and their compounds:

Main outcome: the learner should be able to compare and contrast the chemical properties of the Group 16 elements and their compounds in relation to their position in the Periodic Table.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Trends of chemical properties of Group 16 elements and their compounds: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Trends of chemical properties of Group 16 elements and their compounds.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends of chemical properties of Group 16 elements and their compounds: Outcome Check

Outcome check for Trends of chemical properties of Group 16 elements and their compounds: The learner should be able to compare and contrast the chemical properties of the Group 16 elements and their compounds in relation to their position in the Periodic Table.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends of chemical properties of Group 16 elements and their compounds: Outcome Check

Outcome check for Trends of chemical properties of Group 16 elements and their compounds: Describe the physical properties of the Group 16 elements.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends of chemical properties of Group 17 elements and their compounds: Lesson Opener

Learning focus: Trends of chemical properties of Group 17 elements and their compounds

Country link: a safe Rwandan observation, model, data table, or investigation for Trends of chemical properties of Group 17 elements and their compounds.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the compare and contrast the chemical properties of the Group 17 elements and their compounds in relation to their position in the Periodic Table
- state the natural occurrence of halogens
- describe the extraction methods of halogens
- explain the trends of physical and chemical properties of the Group 17 elements down the group
- describe the trends in strength acidity, volatility and reducing power of halogen hydrides
- describe the chemical properties of chlorates, iodates,

Inquiry questions:

- How can trends of chemical properties of group 17 elements and their compounds help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends of chemical properties of Group 17 elements and their compounds: Learn

Trends of chemical properties of Group 17 elements and their compounds is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Trends of chemical properties of Group 17 elements and their compounds
- Trends
- chemical
- properties
- Group
- elements
- their
- compounds

Trends of chemical properties of Group 17 elements and their compounds: Worked Example

Investigation model for Trends of chemical properties of Group 17 elements and their compounds:
Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Trends of chemical properties of Group 17 elements and their compounds: Vocabulary And Study Skill

Use these words and study moves before you practise Trends of chemical properties of Group 17 elements and their compounds. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Trends of chemical properties of Group 17 elements and their compounds
- Trends
- chemical
- properties
- Group
- elements
- their
- compounds

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Trends of chemical properties of Group 17 elements and their compounds without a clear example, method, or evidence.

Trends of chemical properties of Group 17 elements and their compounds: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Trends of chemical properties of Group 17 elements and their compounds.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Trends of chemical properties of Group 17 elements and their compounds: Practice And Correction

Practice for Trends of chemical properties of Group 17 elements and their compounds:

Main outcome: the compare and contrast the chemical properties of the Group 17 elements and their compounds in relation to their position in the Periodic Table.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Trends of chemical properties of Group 17 elements and their compounds: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Trends of chemical properties of Group 17 elements and their compounds.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends of chemical properties of Group 17 elements and their compounds: Outcome Check

Outcome check for Trends of chemical properties of Group 17 elements and their compounds: The compare and contrast the chemical properties of the Group 17 elements and their compounds in relation to their position in the Periodic Table.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends of chemical properties of Group 17 elements and their compounds: Outcome Check

Outcome check for Trends of chemical properties of Group 17 elements and their compounds: State the natural occurrence of halogens.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Properties and uses of Group 18 elements: Lesson Opener

Learning focus: Properties and uses of Group 18 elements

Country link: a safe Rwandan observation, model, data table, or investigation for Properties and uses of Group 18 elements.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the learner should be able to compare and contrast the properties of the Group 18 elements in relation to their position in the Periodic Table
- state the physical properties of the Group 18 elements
- explain the lack of reactivity of the Group 18 elements
- state the uses of the Group 18 elements
- associate chemical inertia of the Group 18 elements to their full valence shell
- recognise importance of noble gases or Group 18 elements in the daily life

Inquiry questions:

- How can properties and uses of group 18 elements help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Properties and uses of Group 18 elements: Learn

Properties and uses of Group 18 elements is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Properties and uses of Group 18 elements
- Properties
- uses
- Group
- elements
- concept
- observation
- evidence

Properties and uses of Group 18 elements: Worked Example

Investigation model for Properties and uses of Group 18 elements: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Properties and uses of Group 18 elements: Vocabulary And Study Skill

Use these words and study moves before you practise Properties and uses of Group 18 elements. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Properties and uses of Group 18 elements
- Properties
- uses
- Group
- elements
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Properties and uses of Group 18 elements without a clear example, method, or evidence.

Properties and uses of Group 18 elements: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Properties and uses of Group 18 elements.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Properties and uses of Group 18 elements: Practice And Correction

Practice for Properties and uses of Group 18 elements:

Main outcome: the learner should be able to compare and contrast the properties of the Group 18 elements in relation to their position in the Periodic Table.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Properties and uses of Group 18 elements: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Properties and uses of Group 18 elements.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Properties and uses of Group 18 elements: Outcome Check

Outcome check for Properties and uses of Group 18 elements: The learner should be able to compare and contrast the properties of the Group 18 elements in relation to their position in the Periodic Table.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Properties and uses of Group 18 elements: Outcome Check

Outcome check for Properties and uses of Group 18 elements: State the physical properties of the Group 18 elements.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of: Lesson Opener

Learning focus: Trends in chemical properties of

Country link: a safe Rwandan observation, model, data table, or investigation for Trends in chemical properties of.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the compare and contrast the properties of the Period 3 elements and their compounds in relation to their positions in the Periodic Table
- compare the physical properties of the Period 3 elements
- describe the nature of the oxides of the Period 3 elements and the type of bonding in their chlorides, oxides and hydrides
- relate the physical properties of the Period 3 elements to their position in Periodic Table
- relate the physical properties of compounds of the Period 3 elements to their nature of bonds across the period
- develop the sense of analysis while comparing the properties of elements and compounds across the period

Inquiry questions:

- How can trends in chemical properties of help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of: Learn

Trends in chemical properties of explains why atoms join and how bonds affect properties.

Core idea:

- Ionic bonding happens when electrons transfer, often between metals and non-metals.
- Covalent bonding happens when atoms share electrons, usually between non-metals.
- Metallic bonding has metal ions in a sea of delocalized electrons.
- Bonding helps explain melting point, conductivity, solubility, and hardness.
- Dot-and-cross or particle diagrams make bonding easier to compare.

Key words:

- Trends in chemical properties of
- Trends
- chemical
- properties
- concept
- observation
- evidence
- safety

Trends in chemical properties of: Worked Example

Model for Trends in chemical properties of: Compare sodium chloride and a simple covalent molecule.

1. Sodium transfers one electron to chlorine, forming ions.
2. Opposite charges attract, so an ionic bond forms.
3. In a covalent molecule, atoms share electrons instead of transferring them.
4. Ionic compounds often conduct electricity when molten or dissolved; simple covalent substances usually do not.

Conclusion: the type of bond helps explain properties.

Trends in chemical properties of: Vocabulary And Study Skill

Use these words and study moves before you practise Trends in chemical properties of. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Trends in chemical properties of
- Trends
- chemical
- properties
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Trends in chemical properties of without a clear example, method, or evidence.

Trends in chemical properties of: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Trends in chemical properties of.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Trends in chemical properties of: Practice And Correction

Practice for Trends in chemical properties of:

Main outcome: the compare and contrast the properties of the Period 3 elements and their compounds in relation to their positions in the Periodic Table.

1. Complete a table comparing ionic, covalent, and metallic bonding.
2. Draw a simple electron-transfer or sharing model.
3. Match each bond type with one property such as conductivity, melting point, or hardness.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Trends in chemical properties of: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Trends in chemical properties of.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of: Outcome Check

Outcome check for Trends in chemical properties of: The compare and contrast the properties of the Period 3 elements and their compounds in relation to their positions in the Periodic Table.

1. Compare ionic and covalent bonding in one table.
2. State whether electrons are transferred or shared.
3. Link one bond type to one property.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in chemical properties of: Outcome Check

Outcome check for Trends in chemical properties of: Compare the physical properties of the Period 3 elements.

1. Compare ionic and covalent bonding in one table.
2. State whether electrons are transferred or shared.
3. Link one bond type to one property.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Factors that affect chemical equilibrium: Lesson Opener

Learning focus: Factors that affect chemical equilibrium

Country link: a safe Rwandan observation, model, data table, or investigation for Factors that affect chemical equilibrium.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the deduce how concentration, pressure, catalyst and temperature affect the chemical processes in industry
- distinguish between complete and reversible reactions
- explain dynamic equilibrium
- state the characteristics of dynamic equilibrium
- apply Le Chatelier's principle to explain the effects of changes in the temperature, concentration and pressure on a system in equilibrium
- compare and contrast

Inquiry questions:

- How can factors that affect chemical equilibrium help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Factors that affect chemical equilibrium: Learn

Factors that affect chemical equilibrium is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Factors that affect chemical equilibrium
- Factors
- that
- affect
- chemical
- equilibrium
- concept
- observation

Factors that affect chemical equilibrium: Worked Example

Investigation model for Factors that affect chemical equilibrium: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Factors that affect chemical equilibrium: Vocabulary And Study Skill

Use these words and study moves before you practise Factors that affect chemical equilibrium. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Factors that affect chemical equilibrium
- Factors
- that
- affect
- chemical

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Factors that affect chemical equilibrium without a clear example, method, or evidence.

Factors that affect chemical equilibrium: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Factors that affect chemical equilibrium.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Factors that affect chemical equilibrium: Practice And Correction

Practice for Factors that affect chemical equilibrium:

Main outcome: the deduce how concentration, pressure, catalyst and temperature affect the chemical processes in industry.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Factors that affect chemical equilibrium: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Factors that affect chemical equilibrium.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Factors that affect chemical equilibrium: Outcome Check

Outcome check for Factors that affect chemical equilibrium: The deduce how concentration, pressure, catalyst and temperature affect the chemical processes in industry.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Factors that affect chemical equilibrium: Outcome Check

Outcome check for Factors that affect chemical equilibrium: Distinguish between complete and reversible reactions.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Acids and bases: Lesson Opener

Learning focus: Acids and bases

Country link: a safe Rwandan observation, model, data table, or investigation for Acids and bases.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the explain the acid-base theories (Arrhenius, Bronsted-Lowry, Lewis)
- explain the acids and bases using different theories
- explain the differences in behaviour of strong and weak acids and bases using Bronsted-Lowry theory
- classify the acids and bases as strong and weak, depending on their dissociation in aqueous solutions
- distinguish between Lewis and Bronsted- Lowry theory of acids
- write the dissociation of acids and bases and identify acid-base conjugate pairs

Inquiry questions:

- How can acids and bases help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Acids and bases: Learn

Acids and bases is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Acids and bases
- Acids
- bases
- concept
- observation
- evidence
- safety
- conclusion

Acids and bases: Worked Example

Investigation model for Acids and bases: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Acids and bases: Vocabulary And Study Skill

Use these words and study moves before you practise Acids and bases. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Acids and bases
- Acids
- bases
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Acids and bases without a clear example, method, or evidence.

Acids and bases: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Acids and bases.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Acids and bases: Practice And Correction

Practice for Acids and bases:

Main outcome: the explain the acid-base theories (Arrhenius, Bronsted-Lowry, Lewis).

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Acids and bases: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Acids and bases.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Acids and bases: Outcome Check

Outcome check for Acids and bases: The explain the acid-base theories (Arrhenius, Bronsted-Lowry, Lewis).

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Acids and bases: Outcome Check

Outcome check for Acids and bases: Explain the acids and bases using different theories.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Reduction and oxidation reactions: Lesson Opener

Learning focus: Reduction and oxidation reactions

Country link: a safe Rwandan observation, model, data table, or investigation for Reduction and oxidation reactions.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the explain the concept of reduction and oxidation and balance equations for redox reactions
- explain the redox reactions in terms of electron transfer and changes in oxidation state (number)
- explain the concept of disproportionation
- differentiate the reducing agent from the oxidising agent in a redox reaction
- work out the oxidation numbers of elements in the compounds
- perform simple displacement reactions to order elements in terms of oxidising or reducing ability

Inquiry questions:

- How can reduction and oxidation reactions help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Reduction and oxidation reactions: Learn

Reduction and oxidation reactions is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Reduction and oxidation reactions
- Reduction
- oxidation
- reactions
- concept
- observation
- evidence
- safety

Reduction and oxidation reactions: Worked Example

Investigation model for Reduction and oxidation reactions: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Reduction and oxidation reactions: Vocabulary And Study Skill

Use these words and study moves before you practise Reduction and oxidation reactions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Reduction and oxidation reactions
- Reduction
- oxidation
- reactions
- concept

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Reduction and oxidation reactions without a clear example, method, or evidence.

Reduction and oxidation reactions: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Reduction and oxidation reactions.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Reduction and oxidation reactions: Practice And Correction

Practice for Reduction and oxidation reactions:

Main outcome: the explain the concept of reduction and oxidation and balance equations for redox reactions.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Reduction and oxidation reactions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Reduction and oxidation reactions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Reduction and oxidation reactions: Outcome Check

Outcome check for Reduction and oxidation reactions: The explain the concept of reduction and oxidation and balance equations for redox reactions.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Reduction and oxidation reactions: Outcome Check

Outcome check for Reduction and oxidation reactions: Explain the redox reactions in terms of electron transfer and changes in oxidation state (number).

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Energy changes and energy profile diagrams for chemical reactions: Lesson Opener

Learning focus: Energy changes and energy profile diagrams for chemical reactions

Country link: a safe Rwandan observation, model, data table, or investigation for Energy changes and energy profile diagrams for chemical reactions.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- the explain the concept of energy changes and energy profile diagrams for the exothermic and endothermic reactions
- define the term thermochemistry
- state the first law of thermodynamics
- define standard enthalpy of reactions (formation, combustion, neutralisation, and atomisation)
- explain the differences between exothermic and endothermic reactions using energy profile diagrams
- interpret experimental results about energy changes during chemical reactions

Inquiry questions:

- How can energy changes and energy profile diagrams for chemical reactions help you solve a real problem in Chemistry?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Energy changes and energy profile diagrams for chemical reactions: Learn

Energy changes and energy profile diagrams for chemical reactions connects work, power, and energy changes.

Core idea:

- Work is done when a force moves an object through a distance.
- Energy is the ability to do work.
- Power describes how quickly work is done or energy is transferred.
- Energy can change form, such as chemical to movement, electrical to light, or potential to kinetic.

Key words:

- Energy changes and energy profile diagrams for chemical reactions
- Energy
- changes
- energy
- profile
- diagrams
- chemical
- reactions

Energy changes and energy profile diagrams for chemical reactions: Worked Example

Worked example for Energy changes and energy profile diagrams for chemical reactions: A learner lifts a book from the floor to a desk.

1. The learner applies an upward force.
2. The book moves upward through a distance.
3. Work is done on the book.
4. The book gains gravitational potential energy.

Conclusion: work can transfer energy to an object.

Energy changes and energy profile diagrams for chemical reactions: Vocabulary And Study Skill

Use these words and study moves before you practise Energy changes and energy profile diagrams for chemical reactions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Energy changes and energy profile diagrams for chemical reactions
- Energy
- changes
- energy
- profile
- diagrams
- chemical
- reactions

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Energy changes and energy profile diagrams for chemical reactions without a clear example, method, or evidence.

Energy changes and energy profile diagrams for chemical reactions: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Energy changes and energy profile diagrams for chemical reactions.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Energy changes and energy profile diagrams for chemical reactions: Practice And Correction

Practice for Energy changes and energy profile diagrams for chemical reactions:

Main outcome: the explain the concept of energy changes and energy profile diagrams for the exothermic and endothermic reactions.

1. Give two examples of energy transformation.
2. Explain when work is done on an object.
3. Compare work and power in one sentence.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Energy changes and energy profile diagrams for chemical reactions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Energy changes and energy profile diagrams for chemical reactions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Energy changes and energy profile diagrams for chemical reactions: Outcome Check

Outcome check for Energy changes and energy profile diagrams for chemical reactions: The explain the concept of energy changes and energy profile diagrams for the exothermic and endothermic reactions.

1. Give one example of energy changing form.
2. Explain when work is done.
3. Compare work and power in one sentence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Energy changes and energy profile diagrams for chemical reactions: Outcome Check

Outcome check for Energy changes and energy profile diagrams for chemical reactions: Define the term thermochemistry.

1. Give one example of energy changing form.
2. Explain when work is done.
3. Compare work and power in one sentence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Structure of an atom and mass spectrum: Review Clinic 1

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from Structure of an atom and mass spectrum that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Electron configurations of atoms and ions: Review Clinic 2

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from Electron configurations of atoms and ions that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Formation of ionic and metallic bonds: Review Clinic 3

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from Formation of ionic and metallic bonds that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Covalent bond and molecular structures: Review Clinic 4

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from Covalent bond and molecular structures that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Chemistry.

- Structure of an atom and mass spectrum: Explain this term using an example from the book, your school, or your community.
- Electron configurations of atoms and ions: Explain this term using an example from the book, your school, or your community.
- Formation of ionic and metallic bonds: Explain this term using an example from the book, your school, or your community.
- Covalent bond and molecular structures: Explain this term using an example from the book, your school, or your community.
- Variation in trends of the physical properties: Explain this term using an example from the book, your school, or your community.
- Trends in chemical properties of Group 1 elements and their compounds: Explain this term using an example from the book, your school, or your community.
- Trends in chemical properties of Group 2 elements and their compounds: Explain this term using an example from the book, your school, or your community.
- Trends of chemical properties of Group 13 elements and their compounds: Explain this term using an example from the book, your school, or your community.
- Trends in chemical properties of Group 14 elements and their compounds: Explain this term using an example from the book, your school, or your community.
- Trends in chemical properties of Group 15 elements and their compounds: Explain this term using an example from the book, your school, or your community.
- Trends of chemical properties of Group 16 elements and their compounds: Explain this term using an example from the book, your school, or your community.
- Trends of chemical properties of Group 17 elements and their compounds: Explain this term using an example from the book, your school, or your community.
- Properties and uses of Group 18 elements: Explain this term using an example from the book, your school, or your community.
- Trends in chemical properties of: Explain this term using an example from the book, your school, or your community.
- Factors that affect chemical equilibrium: Explain this term using an example from the book, your school, or your community.
- Acids and bases: Explain this term using an example from the book, your school, or your community.
- Reduction and oxidation reactions: Explain this term using an example from the book, your school, or your community.
- Energy changes and energy profile diagrams for chemical reactions: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Chemistry answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Chemistry that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.